Using Mediant Trees to Transduce Rational Approximations of Real Numbers

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Abstract

Given an oracle for a real number z of sufficient precision, we present three contributions. First, an algorithm that constructs a *mediant tree* of increasingly precise rational approximations to z: starting from a Farey-adjacent bounded pair bracketing z, each iteration applies a weighted mediant step with an optimal integer weight w given by a closed-form formula, yielding a new, tighter bounded pair. Second, an algorithm that converts the mediant tree into a *parity sequence* — a string over $\{\mathsf{O}, \mathsf{E}\}$ encoding the approximation path — and shows that for $z > 1$ this sequence is conjectured to be almost equivalent to the Sturmian sequence with slope $1/(1+z)$, agreeing at every position except possibly the last symbol of each finite prefix. Third, we show how to extend both algorithms to real numbers $z \in (0, 1)$ and $z < 0$ via simple algebraic reductions to the $z > 1$ case.

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1 Introduction

The problem of approximating irrational numbers by rationals is classical. The Stern–Brocot tree and Farey sequences provide a systematic account of best rational approximants, and the continued-fraction expansion of a real number gives its convergents — the sequence of closest rational approximations with bounded denominators.

The present paper is motivated by a specific need that arises in the study of generalised Collatz sequences: to compute, algorithmically, a sequence of increasingly precise rational approximations to the logarithm $\log_h(g)$, where $g > h \in \mathbb{N}$, together with an explicit control over the approximation error at each step. The theory we develop is completely general: the core algorithm handles any real number $z > 1$, and Section 10 shows how to reduce all other real numbers to this case.

Our approach works with pairs of rational estimates that simultaneously bracket z from above and below. We call such a pair a *bounded pair* (Definition 2.4). At each step we construct two new mediants from the current pair, choosing an integer weight w so that the new pair is again bounded, and so that the approximation errors strictly decrease. The central contribution is a closed-form formula for the optimal w (Theorem 4.1), which enables efficient iteration.

We also observe that the entire construction can be encoded as a string of symbols from the two-letter alphabet $\{\mathsf{O}, \mathsf{E}\}$, which we call the *parity sequence* (Section 6). This representation connects rational approximation theory to combinatorics on words and, in the Collatz setting, to the odd/even parity structure of trajectories.

A striking consequence (Section 9) is that the parity sequence produced by the mediant tree appears to be identical to the Sturmian sequence with slope $1/(1 + z)$ (Conjecture 9.1, verified computationally for several irrational z). The mediant tree thus provides a compressed transducer for the Sturmian sequence: it encodes a prefix of length N using only $O(\log N)$ real-arithmetic operations, after which the symbols can be enumerated by a purely integer recursive traversal of the tree.

The algorithm requires an oracle for z — a means of evaluating $\lceil oz \rceil$ to sufficient precision at each step — but is otherwise entirely general. Given such an oracle, the mediant tree can represent *any* real number as a parity sequence over $\{\mathsf{O}, \mathsf{E}\}$, making it a universal encoder from reals to binary sequences.

Organisation. Section 2 introduces the core definitions. Section 3 develops the mediant construction and the straddling inequality. Section 4 derives the optimal weight formula. Section 6 introduces parity sequences. Section 7 analyses the rate of convergence. Section 8

presents examples and computational results. Section 10 extends the construction to all real $z \notin \{0, 1\}$. Section 11 concludes with open questions.

2 Definitions and Notation

Throughout Sections 2–9, let $z > 1$ be a fixed real number. Section 10 handles all other real numbers by reduction to this case. We write $\mathbb{N}$ for the positive integers and $\lfloor \cdot \rfloor$, $\lceil \cdot \rceil$ for floor and ceiling.

Definition 2.1 (Estimate)**.** A *rational estimate* of z is any pair $(e, o) \in \mathbb{N}^2$; we regard the pair as representing the rational number e/o.

Definition 2.2 (Good estimate)**.** An estimate e/o is a *good estimate* of z if

$$e \in \{\lfloor o \cdot z \rfloor, \lceil o \cdot z \rceil\},$$

i.e. e/o is one of the two nearest rationals to z with denominator o. A good estimate with $e = \lceil oz \rceil$ satisfies $e/o \geq z$ and is an *upper bound*; one with $e = \lfloor oz \rfloor$ satisfies $e/o \leq z$ and is a *lower bound*.

Remark 2.3. In the special case $z = \log_h(g)$ with $g, h \in \mathbb{N}$, the upper-bound condition $e = \lceil oz \rceil$ translates to the integer inequality $h^{e-1} < g^o \leq h^e$, which is the standard condition on even-step/odd-step counts in Collatz analysis.

Definition 2.4 (Bounded pair)**.** Two good estimates e_l/o_l and e_u/o_u form a *bounded pair* if e_l/o_l is a lower bound, e_u/o_u is an upper bound,

$$\frac{e_l}{o_l} < z \leq \frac{e_u}{o_u},$$

and they satisfy the *Farey adjacency condition*

$$e_u \cdot o_l - e_l \cdot o_u = 1.$$

The Farey adjacency condition ensures that the pair lies on adjacent branches of the Stern–Brocot tree and that no rational with smaller denominator lies strictly between e_l/o_l and e_u/o_u.

Definition 2.5 (λ and ε)**.** For an estimate e/o of z, define the *relative ratio*

$$\lambda = \frac{e/o}{z}$$

and the *relative error*

$$\varepsilon = \lambda - 1 = \frac{e/o}{z} - 1.$$

An upper-bound good estimate satisfies $\lambda \geq 1$ (so $\varepsilon \geq 0$); a lower-bound good estimate satisfies $\lambda \leq 1$ (so $\varepsilon \leq 0$). In both cases $|\varepsilon|$ is small when o is large.

We will index estimates by $i, i+1, j, k$ and write λ_i, ε_i, etc. for the corresponding values. In a bounded pair we always orient so that $|\varepsilon_i| > |\varepsilon_{i+1}|$, i.e. r_{i+1} is the better estimate of the two.

Definition 2.6 (Δ and K)**.** Given a bounded pair $(r_i, r_{i+1}) = (e_i/o_i,\ e_{i+1}/o_{i+1})$ with $|\varepsilon_i| > |\varepsilon_{i+1}|$, define

$$\Delta = e_{i+1}o_i - e_i o_{i+1}$$

and

$$K = o_{i+1}(e_{i+1} - z \cdot o_{i+1}).$$

The Farey adjacency condition in Definition 2.4 requires $e_u \cdot o_l - e_l \cdot o_u = 1$, which is precisely $|\Delta| = 1$ (the sign depends on which of r_i, r_{i+1} is the upper bound). One checks that $|K| \leq 1$ for any good estimate with $|\varepsilon_{i+1}|$ small, and that $\operatorname{sgn}(K) = \operatorname{sgn}(\Delta)$ throughout the iteration.

Domain restriction. The definitions in this section assume $z > 1$. Section 10 shows how to reduce all other real numbers to this case.

3 The Mediant Construction

The classical *mediant* of two fractions e_i/o_i and e_{i+1}/o_{i+1} is $(e_i + e_{i+1})/(o_i + o_{i+1})$. We generalise this to a weighted family parametrised by a positive integer w.

Definition 3.1 (Weighted mediants j and k)**.** Given a bounded pair (r_i, r_{i+1}) and a positive integer $w \geq 1$, define

$$r_j = \frac{e_i + (w-1)\, e_{i+1}}{o_i + (w-1)\, o_{i+1}},$$

$$r_k = \frac{e_i + w\, e_{i+1}}{o_i + w\, o_{i+1}}.$$

Theorem 3.2 (Mediant convergence)**.** *Let (r_i, r_{i+1}) be a bounded pair for z with $|\varepsilon_i| > |\varepsilon_{i+1}|$, and let $w \geq 1$ be a positive integer such that (r_j, r_k) brackets z (i.e. $(\lambda_j - 1)(\lambda_k - 1) \leq 0$). Then*

$$|\varepsilon_j| \leq |\varepsilon_i| \qquad \textit{and} \qquad |\varepsilon_k| < |\varepsilon_{i+1}| < |\varepsilon_i|.$$

(When $w = 1$, $r_j = r_i$ exactly so $|\varepsilon_j| = |\varepsilon_i|$; otherwise $|\varepsilon_j| < |\varepsilon_i|$.) Moreover (r_j, r_k) is itself a bounded pair. The two mediants are assigned as r_{i+2} and r_{i+3}, where r_{i+3} is whichever of r_j, r_k has the smaller $|\varepsilon|$ (the better estimate), and r_{i+2} takes the other. Since r_k always satisfies $|\varepsilon_k| < |\varepsilon_{i+1}|$, we have $|\varepsilon_{i+3}| < |\varepsilon_{i+1}|$ regardless of which candidate is assigned to r_{i+3}.

Proof. The denominators satisfy $o_j = o_i + (w-1)o_{i+1} < o_i + w\, o_{i+1} = o_k$, so r_j and r_k are distinct. The ratio λ_j is a weighted combination of λ_i and λ_{i+1}:

$$\lambda_j = \frac{o_i\, \lambda_i + (w-1)o_{i+1}\, \lambda_{i+1}}{o_i + (w-1)o_{i+1}}.$$

When $w = 1$ this gives $\lambda_j = \lambda_i$ exactly, so $|\varepsilon_j| = |\varepsilon_i|$. When $w \geq 2$, λ_j is a strict convex combination of λ_i and λ_{i+1}, lying strictly between them, so $|\lambda_j - 1| < |\lambda_i - 1|$, i.e. $|\varepsilon_j| < |\varepsilon_i|$. For r_k (weight $w \geq 1$), the combination is always strict, so $|\lambda_k - 1| < |\lambda_i - 1|$ and $|\lambda_k - 1| < |\lambda_{i+1} - 1|$, i.e. $|\varepsilon_k| < |\varepsilon_{i+1}| < |\varepsilon_i|$. The Farey adjacency condition for (r_j, r_k) follows by direct expansion:

$$\begin{aligned} e_k o_j - e_j o_k &= (e_i + w\, e_{i+1})(o_i + (w-1)o_{i+1}) - (e_i + (w-1)e_{i+1})(o_i + w\, o_{i+1}) \\ &= \big[(w-1) - w\big] e_i\, o_{i+1} + \big[w - (w-1)\big] e_{i+1}\, o_i \\ &= -e_i\, o_{i+1} + e_{i+1}\, o_i = \Delta, \end{aligned}$$

so $|e_k o_j - e_j o_k| = |\Delta| = 1$, and the Farey adjacency of (r_j, r_k) is inherited directly from that of (r_i, r_{i+1}). □

Corollary 3.3 (Straddling inequality)**.** *Under the conditions of Theorem 3.2,*

$$\varepsilon_j\, \varepsilon_k < 0 \qquad \textit{and} \qquad |\varepsilon_j||\varepsilon_k| < |\varepsilon_i|\, |\varepsilon_{i+1}|.$$

Proof. The first inequality holds because r_j and r_k straddle z, so their errors have opposite signs. For the second: $|\varepsilon_j| \leq |\varepsilon_i|$ and $|\varepsilon_k| < |\varepsilon_{i+1}|$ (Theorem 3.2), so $|\varepsilon_j||\varepsilon_k| \leq |\varepsilon_i||\varepsilon_k| < |\varepsilon_i|\, |\varepsilon_{i+1}|$. □

4 The Optimal Weight Formula

A key question is: for which integer w is the resulting bounded pair (r_j, r_k) the most accurate, i.e. which w minimises the larger of $|\varepsilon_j|$ and $|\varepsilon_k|$?

The real-valued optimum is found by differentiating the left-hand side of the straddling inequality with respect to w and setting the derivative to zero. Recall that $r_j = \text{Mediant}(r_i, r_{i+1}, w-1)$ and $r_k = \text{Mediant}(r_i, r_{i+1}, w)$, so $\lambda_j - 1 = \varepsilon_j$ and $\lambda_k - 1 = \varepsilon_k$ are functions of w. Computing the derivative gives

$$\frac{d}{dw}\text{LHS} = \frac{\Delta}{z}\left[\frac{\varepsilon_j}{(o_i + w\, o_{i+1})^2} + \frac{\varepsilon_k}{(o_i + (w-1)o_{i+1})^2}\right].$$

Setting this to zero and solving yields the following closed-form expression.

Theorem 4.1 (Optimal weight)**.** *The real-valued minimiser of the combined approximation error is*

$$w = \frac{1}{2} - \frac{o_i}{o_{i+1}} + \frac{\Delta \pm \sqrt{\Delta^2 - K^2}}{2K},$$

where

$$\Delta = e_{i+1} o_i - e_i o_{i+1}, \qquad K = o_{i+1}(e_{i+1} - z \cdot o_{i+1}).$$

The formula yields two real candidates (one for +, one for −), and the integer value of w is found by taking the floor or ceiling of each candidate and selecting the one that satisfies $(\lambda_j - 1)(\lambda_k - 1) \leq 0$ (i.e. that the pair brackets z). A full derivation is given in Appendix A.

Remark 4.2. When $K = 0$ the formula is undefined, which corresponds to the situation where e_{i+1}/o_{i+1} is an exact rational representation of z and no further improvement is possible. When $|K|$ is small, the discriminant $\Delta^2 - K^2 \approx 1$ and the two candidates are approximately $1/(1-K)$ apart, so the optimal w is of moderate size.

Algorithm. The iteration proceeds as follows.

1. Start with an initial bounded pair (r_0, r_1) for z.
2. Compute Δ and K.
3. If $K = 0$, terminate.
4. Evaluate the two candidates for w from Theorem 4.1; search over $\lfloor w_- \rfloor$ and $\lceil w_+ \rceil$ for the integer $w \geq 1$ that satisfies the bracketing condition.
5. Form the new bounded pair (r_j, r_k) and repeat from step 2.

Because o_{i+1} grows monotonically along the iteration, the denominator of K grows and the iteration converges (Section 7).

5 The Mediant Tree

Each estimate produced by the algorithm is either a *leaf* — one of the integers $\lfloor z \rfloor$ or $\lceil z \rceil$ — or a *Mediant node* built recursively from two sub-estimates. This recursive structure defines a binary tree for each estimate, which we call its *mediant tree*.

Definition 5.1 (Mediant tree)**.** The *mediant tree* of an estimate r is defined recursively:

- If $r \in \{\lfloor z \rfloor, \lceil z \rceil\}$, then r is a *leaf node*.
- Otherwise r is an *interior node* Mediant(r_0, r_1, w) with left subtree r_0, right subtree r_1, and integer weight $w \geq 0$, representing the rational

$$r = \frac{e_0 + w\, e_1}{o_0 + w\, o_1}.$$

The two leaves are the unique consecutive integers bracketing z: $\lfloor z \rfloor < z \leq \lceil z \rceil$, forming the initial bounded pair $(r_0, r_1) = (\lfloor z \rfloor, \lceil z \rceil)$ of the algorithm. For $z = \log_2 3 \approx 1.585$ the leaves are 1 and 2.

Each bounded pair (r_{2n}, r_{2n+1}) in the algorithm's sequence therefore consists of two mediant trees that share the same leaves. As n increases the trees grow deeper, recording the full history of mediant constructions that led to each estimate.

Example 5.2 (Mediant trees for $\log_2 3$)**.** At step $n = 1$ the bounded pair (3/2, 5/3) has trees:

$$r_2 = 3/2 = \text{Mediant}(\underbrace{1}_{\text{leaf}}, \underbrace{2}_{\text{leaf}}, \ w = 1),$$
$$r_3 = 5/3 = \text{Mediant}(\underbrace{1}_{\text{leaf}}, \underbrace{2}_{\text{leaf}}, \ w = 2).$$

At step $n = 2$ the pair (3/2, 8/5) has trees:

$$r_4 = 3/2 = \text{Mediant}(\text{Mediant}(1,\, 2,\, 1),\ \text{Mediant}(1,\, 2,\, 2),\ w = 0),$$
$$r_5 = 8/5 = \text{Mediant}(\text{Mediant}(1,\, 2,\, 1),\ \text{Mediant}(1,\, 2,\, 2),\ w = 1).$$

Every internal node carries a weight $w \geq 0$; a weight of 0 means the left subtree is used directly (contributing zero copies of the right subtree).

Operational use. The mediant tree is the practical device for generating parity sequences of arbitrary length and precision. Given a target z and a desired precision, one runs the algorithm to depth n, obtaining a bounded pair (r_{2n}, r_{2n+1}) whose mediant trees encode parity sequences of length $e_{2n+1} + o_{2n+1}$. The parity sequence $\pi(r_{2n+1})$ (Definition 6.2) is then read off the tree by the recursive rule: an interior node $\text{Mediant}(r_0, r_1, w)$ with $\Delta > 0$ yields $\pi(r_1)^w \cdot \pi(r_0)$, and with $\Delta < 0$ yields $\pi(r_0) \cdot \pi(r_1)^w$. Deeper trees produce longer prefixes of the Sturmian sequence with slope $1/(z+1)$ (Section 9).

6 Parity Sequences

The mediant construction admits a compact combinatorial encoding.

Definition 6.1 (Parity alphabet and base word)**.** Let $\{\mathsf{O}, \mathsf{E}\}^*$ denote the set of finite strings over the two-letter alphabet $\{\mathsf{O}, \mathsf{E}\}$. For a base estimate $e/1$ (denominator 1) define its *parity sequence*

$$\pi(e/1) = \underbrace{\mathsf{E} \cdots \mathsf{E}}_{e} \mathsf{O},$$

a string of e copies of E followed by one O.

Definition 6.2 (Parity sequence of a mediant)**.** Given a sequence of estimates $r_0, r_1, r_2, \ldots$ produced by the algorithm, the parity sequences of the next pair (r_{i+2}, r_{i+3}) are computed from the bounded pair (r_i, r_{i+1}) and weight w_{i+3}, where r_{i+3} is the better estimate and r_{i+2} the other.

- If $\Delta > 0$ (so r_{i+1} is the upper bound, $r_{i+1} \geq z$), $\pi(r_i)$ appears as a suffix:

$$\pi(r_{i+2}) = \pi(r_{i+1})^{w_{i+3}-1} \cdot \pi(r_i),$$
$$\pi(r_{i+3}) = \pi(r_{i+1})^{w_{i+3}} \cdot \pi(r_i).$$

- If $\Delta < 0$ (so r_{i+1} is the lower bound, $r_{i+1} < z$), $\pi(r_i)$ appears as a prefix:

$$\pi(r_{i+2}) = \pi(r_i) \cdot \pi(r_{i+1})^{w_{i+3}-1},$$
$$\pi(r_{i+3}) = \pi(r_i) \cdot \pi(r_{i+1})^{w_{i+3}}.$$

Proposition 6.3 (Parity sequence encodes the approximation path)**.** *Any consecutive pair $(\pi(r_{i+2}), \pi(r_{i+3}))$ in the sequence uniquely determines the initial pair (r_0, r_1) and the sequence of weights $w_2, w_3, \ldots$ used at each step.*

Proof sketch. The lengths satisfy $|\pi(r_n)| = e_n + o_n$. Given the pair $(\pi(r_{i+2}), \pi(r_{i+3}))$, the recovery proceeds as follows.

1. $\pi(r_i)$ is recovered by trying both cases: if the pair shares a non-trivial longest common suffix then $\Delta > 0$ and $\pi(r_i)$ is that suffix; if it shares a non-trivial longest common prefix then $\Delta < 0$ and $\pi(r_i)$ is that prefix. Exactly one case yields a consistent decomposition per Definition 6.2, and this simultaneously recovers $\pi(r_i)$ and the sign of Δ.

2. $|\pi(r_{i+1})| = |\pi(r_{i+3})| - |\pi(r_{i+2})|$, since the two words differ by exactly one copy of $\pi(r_{i+1})$.

3. $w_{i+3} = (|\pi(r_{i+3})| - |\pi(r_i)|) \,/\, |\pi(r_{i+1})|$ exactly.

4. $\pi(r_{i+1})$ is extracted as the remaining portion of $\pi(r_{i+3})$ after removing $\pi(r_i)$, then divided into w_{i+3} equal copies.

Applying this recursively mirrors the Euclidean algorithm, uniquely recovering the initial pair (r_0, r_1) and all weights. □

Example 6.4 (Parity sequence for $\log_2 3$)**.** Starting from the initial bounded pair $(r_0, r_1) = (1/1,\, 2/1)$ with $z = \log_2 3 \approx 1.58496$, the algorithm produces the following sequence of bounded pairs. At each step n, weight w_{2n+1} is used to compute r_j and r_k. The better of the two (smaller $|\varepsilon|$) is assigned to r_{2n+3} and the other to r_{2n+2}, forming the next bounded pair; r_{2n+1} is always the better estimate.

n	r_{2n}	r_{2n+1}	$\varepsilon_{r_{2n}}$	$\varepsilon_{r_{2n+1}}$	w_{2n+1}	r_j	r_k
0	1/1	2/1	−0.369	+0.262	2	3/2	5/3
1	3/2	5/3	−0.054	+0.052	1	3/2	8/5
2	3/2	8/5	−0.054	+0.010	3	19/12	27/17
3	27/17	19/12	+0.002	−0.001	3	65/41	84/53

The parity sequence of $r_7 = 19/12$ (length $19 + 12 = 31$) decomposes recursively. The leaf words in Sturmian order are $\pi(r_0) = \texttt{EO}$ and $\pi(r_1) = \texttt{EEO}$. Reading the decomposition bottom-up:

Level 1 ($n = 0$, $w_1 = 2$, $\Delta > 0$ so suffix, $\pi(r_3) = \pi(r_1)^2 \cdot \pi(r_0)$, $\pi(r_2) = \pi(r_1)^1 \cdot \pi(r_0)$):

$$\underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\underbrace{\texttt{EEO}}_{\pi(r_1)}\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_3)=\texttt{EEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_2)=\texttt{EEOEO}}$$

Level 2 ($n = 1$, $w_3 = 1$, $\Delta > 0$ so suffix, $\pi(r_5) = \pi(r_3)^1 \cdot \pi(r_2)$, $\pi(r_4) = \pi(r_2)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEO}}_{\pi(r_3)}\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_5)=\texttt{EEOEEOEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_4)=\texttt{EEOEO}}$$

Level 3 ($n = 2$, $w_5 = 3$, $\Delta > 0$ so suffix, $\pi(r_7) = \pi(r_5)^2 \cdot \pi(r_4)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\ \underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\ \underbrace{\texttt{EEOEO}}_{\pi(r_4)}}_{\pi(r_7)}$$

Remark 6.5. In the Collatz setting ($z = \log_2 3$, $g = 3$, $h = 2$), the symbol O in a parity sequence corresponds to an odd step ($x \mapsto 3x + 1$) and E to an even step ($x \mapsto x/2$). The parity sequence of a good estimate e/o thus describes an idealised Collatz trajectory with o odd steps and e even steps whose ratio e/o approximates $\log_2 3$.

Remark 6.6 (Every prefix encodes the same real number). The parity sequences produced by the mediant tree have an unusual self-consistency property: every prefix of the sequence is itself the parity sequence of some estimate of the same real number z. That is, truncating the sequence at any point yields a shorter parity sequence that still encodes an approximation to z — not some other value.

Moreover, longer sequences encode better approximations on average: as the sequence grows, the estimates r_{2n+1} converge to z, so the ratio e/o of E's to O's in a long prefix approaches z more closely than in a short one.

This monotone improvement holds only at the level of the mediant tree: moving from one layer to the next always yields a strictly better estimate (Theorem 7.1). Within a single layer, however, the two candidates r_j and r_k need not both improve on every previous estimate — one is assigned to r_{i+3} (the better) and the other to r_{i+2} (which may represent a local regression relative to r_{i+3}). The strict improvement guarantee applies across layers, not within them.

7 Convergence Analysis

Theorem 7.1 (Strict decrease of $|\varepsilon|$)**.** *At each iteration of the algorithm, $|\varepsilon_{i+1}|$ strictly decreases.*

Proof. By Theorem 3.2, both $|\varepsilon_j|$ and $|\varepsilon_k|$ are strictly less than $|\varepsilon_{i+1}|$. The next iterate uses the better of r_j and r_k as its r_{i+1}, so $|\varepsilon_{i+1}^{\text{new}}| < |\varepsilon_{i+1}^{\text{old}}|$. □

Proposition 7.2 (Relationship to continued fractions)**.** *The sequence of denominators o_{i+1} generated by the algorithm is a subsequence of the denominators of the convergents of z in its continued-fraction expansion. Specifically, the algorithm recovers the convergents at each step where the optimal w equals the corresponding partial quotient of z.*

Remark 7.3. The convergence of the algorithm is at least as fast as the continued-fraction convergents, since the optimal w formula ensures that no redundant steps are taken. In practice the algorithm achieves superlinear growth of the denominator o_{i+1}, since the optimal w can be large (corresponding to a large partial quotient in the continued-fraction expansion of z).

8 Examples and Computational Results

We illustrate the algorithm on two targets.

8.1 Approximating $\log_2 3$

Table 1 shows the first several bounded pairs produced by the algorithm for $z = \log_2 3$. The initial pair is $r_0 = 1/1$, $r_1 = 2/1$ (r_1 is the better estimate throughout).

Table 1: Successive bounded pairs approximating $\log_2 3 \approx 1.58496\ldots$

n	r_{2n}	r_{2n+1}	$\varepsilon_{r_{2n}}$	$\varepsilon_{r_{2n+1}}$	w_{2n+1}	r_j	r_k
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53
4	65/41	84/53	$+2.5 \times 10^{-4}$	-3.6×10^{-5}	6	485/306	569/359
5	485/306	569/359	$+3.0 \times 10^{-6}$	-2.7×10^{-6}	1	485/306	1054/665
6	485/306	1054/665	$+3.0 \times 10^{-6}$	-6.0×10^{-8}	24	24727/15601	25781/16266

8.2 Approximating π

For comparison, Table 2 shows the first several steps for $z = \pi$, starting from $r_0 = 4/1$, $r_1 = 3/1$. The classical approximant $355/113$ appears at step $n = 2$ with $\varepsilon \approx +8.49 \times 10^{-8}$, after which the weight $w_5 = 293$ produces a large jump in precision in a single step.

Table 2: Successive bounded pairs approximating $\pi \approx 3.14159\ldots$

n	r_{2n}	r_{2n+1}	$\varepsilon_{r_{2n}}$	$\varepsilon_{r_{2n+1}}$	w_{2n+1}	r_j	r_k
0	4/1	3/1	$+0.273$	-0.045	7	22/7	25/8
1	25/8	22/7	-5.0×10^{-3}	$+4.0 \times 10^{-4}$	15	333/106	355/113
2	333/106	355/113	-2.7×10^{-5}	$+8.5 \times 10^{-8}$	293	103993/33102	104348/33215

9 Relationship to Sturmian Sequences

A *Sturmian sequence* over $\{\mathsf{O}, \mathsf{E}\}$ with slope $\alpha \in (0, 1)$ is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \qquad n \geq 0,$$

encoding the sequence as O when $s_n = 1$ and E when $s_n = 0$. Since $\alpha < 1$, we have $s_0 = \lfloor \alpha \rfloor = 0$, so the Sturmian sequence always begins with E.

Conjecture 9.1 (Parity sequences are Sturmian)**.** *Let $z > 1$ be irrational and let $\alpha = 1/(z+1)$. The infinite parity sequence produced by the mediant tree is identical to the Sturmian sequence with slope α.*

For a finite parity sequence of length N (the sequence of estimate r_{2n+1} at depth n), the two sequences agree at every position $0 \leq i < N-1$. At the final position $i = N-1$ they may differ: the mediant sequence always emits O (the terminal symbol of its last primitive word), while the Sturmian sequence may emit E at that position (being mid-word at depth n).

The single terminal discrepancy is an artefact of truncation, not a genuine disagreement: the mediant sequence closes its last primitive word with O at position $N-1$, whereas the Sturmian sequence — which continues indefinitely — may still be in the interior of a longer word at that depth. As the depth increases and $N \to \infty$, both sequences converge to the same infinite word.

This has been verified computationally for several irrational values of z, including $\sqrt{2}$, $\log_2 3$, $\log_2 5$, ϕ, and π. For $z = \log_2 3$ the verification extends to sequences of length up to 27,869,189 symbols: at every depth the two sequences agree everywhere except possibly the last symbol, where the mediant emits O and the Sturmian emits E.

Remark 9.2 (The mediant sequence is the better finite approximation). At depths where the two sequences differ, the mediant's finite parity sequence is a closer approximation to $\alpha = 1/(z+1)$ in terms of running O-density. The terminal O emitted by the mediant sequence gives a density slightly *above* α, whereas the Sturmian's terminal E gives a density substantially *below* α.

This reflects a structural asymmetry in the two methods. The Sturmian algorithm at position $N-1$ is mid-word in a longer primitive word, so it emits E — inherently favouring the above-z estimate and leaving the running density below α. The mediant tree, by contrast, explicitly selects at each step the estimate with smaller $|\varepsilon|$, which is the lower-bound estimate r_{2n} (the below-z member of the pair) when that is the better of the two. Closing the sequence with O at that point reflects this selection: it is the correct terminal for a word whose ratio e/o slightly underestimates z, nudging the running density up toward α.

For $z = \log_2 3$, the improvement at the discrepancy depths is substantial: at $N = 1719$ the mediant density error is 1.4×10^{-8} against the Sturmian's 5.8×10^{-4} — roughly four orders of magnitude better.

The O-density of $\pi(r)$ for an estimate e/o is

$$\frac{o}{e+o} = \frac{1}{e/o+1} \xrightarrow{e/o \to z} \frac{1}{z+1} = \alpha,$$

consistent with the Sturmian slope.

Efficiency and lazy generation. Although the parity sequences are Sturmian, the mediant tree is a far more efficient way to generate them. To produce a prefix of N symbols, the naive Sturmian algorithm must evaluate $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ at each of the N positions, requiring $O(N)$ real-arithmetic operations.

The mediant algorithm instead builds a compressed representation of the same prefix. Each iteration requires a constant number of real-arithmetic operations to compute the next bounded pair and weight; after k iterations the tree encodes a Sturmian prefix of length $N = e_{2k+1} + o_{2k+1}$, which grows exponentially in k. Thus only $O(\log N)$ real-arithmetic operations are needed to encode a prefix of length N.

Once the tree is built, enumerating the N symbols requires *no real arithmetic at all*: each node specifies an integer weight w and two child words, and reading off the sequence is a recursive depth-first traversal governed entirely by integer counts. No auxiliary memory beyond the call stack is required.

OO-avoidance. Since $z > 1$ implies $\alpha = 1/(z+1) < 1/2$, the Sturmian sequence avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. For comparison, when $\alpha > 1/2$ (i.e. $z < 1$) consecutive O's are unavoidable — for example $\alpha = 2/3$ gives $\mathsf{OOEOOE}\cdots$ immediately. The mediant tree gives an independent structural proof: at the leaves each base word is $\mathsf{E}^k\mathsf{O}$ for some $k \geq 1$, and at each interior node exactly one copy of $\pi(r_i)$ is emitted with at least one copy of $\pi(r_{i+1})$ on each side, so $\pi(r_i)$ is never adjacent to itself; by induction OO never arises.

Remark 9.3 (Sturmian sequences are more restricted than general Collatz sequences). A Sturmian sequence with slope $\alpha < 1/2$ consists of exactly two distinct words: $\mathsf{E}^{\lfloor 1/\alpha \rfloor - 1}\mathsf{O}$ and $\mathsf{E}^{\lceil 1/\alpha \rceil - 1}\mathsf{O}$, concatenated in the Sturmian pattern. For slope $\alpha = 1/(z+1)$ these are precisely $\mathsf{E}^{\lfloor z \rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil z \rceil}\mathsf{O}$ — exactly the leaf words of the mediant tree for z. A Sturmian sequence can never contain a word of any other length.

A general Collatz parity sequence is far less constrained: an actual trajectory can produce EO, EEO, EEEO, and so on — E-runs of any length, interleaved freely. The Sturmian sequence is the *ideally balanced* special case, constrained to exactly the two word lengths $\mathsf{E}^{\lfloor z \rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil z \rceil}\mathsf{O}$.

Remark 9.4. The weights w_{2n+1} are related to the partial quotients of the continued fraction expansion of z, though they are not in general identical to them. A large weight at step n corresponds to a large partial quotient, producing a long run of one symbol type in the parity sequence — the same phenomenon that gives rise to the irregular spacing of O's and E's in an irrational Sturmian sequence. Conversely, small weights correspond to small partial quotients, and the parity sequence advances by only a short block at each such step.

Remark 9.5 (The $\log_2 3$ case). For the Collatz application $z = \log_2 3$, the limiting O-density is

$$\frac{1}{z+1} = \frac{1}{\log_2 3 + 1} = \frac{1}{\log_2 6} = \log_6 2 \approx 0.3869.$$

The convergents of $\log_2 3$ are $2/1, 8/5, 19/12, 84/53, \ldots$, and these appear as the r_{2n+1} values in Table 1. In the Collatz context, the O symbols represent odd steps ($x \mapsto 3x+1$) and the E symbols represent even steps ($x \mapsto x/2$). An O-density of $\log_6 2 = 1/\log_2 6$ describes

the *ideally balanced* Collatz parity pattern: the asymptotic ratio of odd to even steps in a trajectory that descends as quickly as theory permits.

Remark 9.6 (The $g = h^2 - 1$ family). For the generalised Collatz system with odd step $x \mapsto gx+1$ and even step $x \mapsto x/h$, the Sturmian slope is $1/(\log_h g+1) = 1/\log_h(gh)$. The special family $g = h^2 - 1$ satisfies $\log_h g \in (1, 2)$ for all $h \geq 2$, so the slope $1/\log_h(gh) \in (1/3, 1/2)$ for every member. The first convergent of $\log_h g$ from above is always $2/1$, corresponding to the parity sequence EEO (two E's, one O) and the minimal cycle $\{h^2, h, 1\}$ shared by every system in this family.

10 Extension to $0 < z < 1$

The algorithm of Sections 3–6 requires $z > 1$. We now show how to handle $z \in (0, 1)$ by algebraic reduction, in two stages.

If one were content to accept OO in the parity sequence, no special treatment would be needed: the algorithm and the Sturmian definition extend formally to any $z > 0$, and for $z < 1$ the slope $\alpha = 1/(1 + z) > 1/2$ simply makes OO unavoidable. The reductions below are necessary precisely because we require parity sequences that are valid Collatz parity sequences — and a Collatz trajectory with two consecutive odd steps is impossible (after an odd step $x \mapsto 3x + 1$ the result is even, so the next step must be even). The reductions map $z \in (0, 1)$ to $z' > 1$, yielding an OO-free sequence whose e and o counts then determine a rational approximant to the original z.

10.1 Case $0 < z < \frac{1}{2}$

For $z \in (0, \frac{1}{2})$, define

$$z' = \frac{1 - z}{z} = \frac{1}{z} - 1 > 1.$$

Run the algorithm on z'. A bounded pair $e/o \approx z'$ satisfies $e/o \approx (1 - z)/z$, so

$$z \approx \frac{o}{e + o}.$$

The approximant to z is therefore $o/(e + o)$, with numerator o (the O-count of the parity sequence) and denominator $e + o$ (its total length).

Parity sequence. The parity sequence for z' has O-density

$$\frac{1}{1 + z'} = \frac{1}{1 + (1 - z)/z} = z,$$

which is less than $\frac{1}{2}$ since $z < \frac{1}{2}$. Consequently the OO-avoidance result of Section 9 applies: the parity sequence for z' is OO-free, and its O-density converges to z.

Sturmian identity. The Sturmian sequence with slope $\alpha' = z$ is the infinite parity sequence produced by the mediant tree for z' (Conjecture 9.1). This Sturmian sequence encodes z directly: the O-density of any long prefix approximates z with the same precision as the underlying mediant approximation of z'.

10.2 Case $\frac{1}{2} < z < 1$

For $z \in (\frac{1}{2}, 1)$, note that $1 - z \in (0, \frac{1}{2})$, so define

$$z' = \frac{z}{1-z} > 1$$

and run the algorithm on z'. A bounded pair $e/o \approx z'$ satisfies $e/o \approx z/(1-z)$, so

$$z \approx \frac{e}{e+o}.$$

No modification of the parity sequence is required: the sequence for z' is used solely to produce the counts e and o, which then yield the approximant $e/(e+o)$ to the original z.

Parity sequence. The parity sequence for z' has O-density

$$\frac{1}{1+z'} = \frac{1}{1+z/(1-z)} = 1 - z < \tfrac{1}{2},$$

so it is OO-free and its Sturmian slope is $1 - z$. The sequence encodes z'; the rational approximant $e/(e+o)$ to z is read off from the E- and O-counts of that sequence.

Summary. Table 3 collects the reductions for all four ranges of z.

Table 3: Reductions to the $z > 1$ algorithm for all ranges of z

Range of z	Reduction z'	Approximant to z	O-density
$z > 1$	$z' = z$	e/o	$1/(1+z) < \frac{1}{2}$
$0 < z < \frac{1}{2}$	$z' = (1-z)/z$	$o/(e+o)$	$z < \frac{1}{2}$
$\frac{1}{2} < z < 1$	$z' = z/(1-z)$	$e/(e+o)$	$1 - z < \frac{1}{2}$
$z < 0$	$z' = -z$ (or above)	negate	—

11 Conclusion and Open Problems

We have presented a theory of rational approximation for real numbers based on bounded pairs and the weighted mediant construction. The core algorithm applies to $z > 1$; all other real numbers are handled by algebraic reduction (Section 10). The central result, Theorem 4.1, gives a closed-form formula for the optimal mediant weight at each step, enabling an efficient iterative algorithm. The parity sequence representation (Section 6) is conjectured to produce exactly the Sturmian sequence with slope $1/(z+1)$ in the limit (Conjecture 9.1), with finite prefixes agreeing everywhere except possibly the terminal symbol; this has been verified computationally but not yet proved. The mediant tree generates exponentially long prefixes per iteration. Section 10 extends the construction to all $z \in (0, 1)$ via the reductions $z' = (1-z)/z$ (for $z < \frac{1}{2}$) and $z' = z/(1-z)$ (for $z > \frac{1}{2}$); in both cases the algorithm runs on the reduced $z' > 1$ and the approximant to z is read off from the resulting e and o counts.

Open problems.

1. *Proof of the Sturmian identity.* Conjecture 9.1 has two parts, both open. First, that finite parity sequences of length N agree with the Sturmian sequence at every position except possibly the last: although verified computationally for several irrational values ($\sqrt{2}$, $\log_2 3$, $\log_2 5$, ϕ, π), with the $\log_2 3$ case checked up to $N = 27{,}869{,}189$, no general proof is known. Second, that the infinite limit of the mediant construction is exactly the Sturmian sequence with slope $1/(z+1)$ for all irrational $z > 1$. The mediant construction builds parity sequences via a morphism-based concatenation rule; the connection to Sturmian morphisms is the natural proof route.

2. *The parent-swap phenomenon.* To track the Sturmian sequence correctly across successive mediant depths, the parent pair (r_i, r_{i+1}) must sometimes be swapped when constructing child mediants — i.e. the roles of the lower and upper bound estimates are exchanged at certain levels of the tree. Computationally, the condition for a swap is clear: it occurs precisely when $\Delta < 0$. What has not been established is *why* $\Delta < 0$ is the necessary and sufficient condition — i.e. a proof that the Sturmian order is preserved if and only if the parent pair is swapped at levels where $\Delta < 0$.

Related Work

Eliahou [1] studies rational approximation of $\log_2 3$ in the context of bounding nontrivial cycle lengths in the Collatz problem. His central rational approximation result (Theorem 3.2) identifies the best upper rational approximants to an irrational θ — those fractions k/l with $\theta < k/l \leq \theta'$ minimising k or l — as the *upper convergents* $p_{n,i}/q_{n,i}$ derived from the continued fraction expansion of θ. Applied to $\theta = \log_2 3$, this yields explicit lower bounds on Collatz cycle lengths.

Our work differs in two essential respects. First, our algorithm never computes a continued fraction expansion: it works directly on bounded pairs via the optimal weight formula (Theorem 4.1), and the weights w_{2n+1} emerge from that construction — related to, but not in general identical to, the continued fraction partial quotients. Both approaches do, however, share a common dependency: each requires an oracle capable of supplying an arbitrary-precision representation of the target value z, sufficient to evaluate $\lceil oz \rceil$ (our algorithm) or $\lfloor (n+1)\alpha \rfloor$ (Sturmian) to the required accuracy. Neither method generates more precision than the oracle provides.

Second, and more fundamentally, Eliahou's framework provides no means of constructing a *parity sequence* realising a given approximant. Knowing that k/l is an upper convergent to $\log_2 3$ says nothing about how to build an integer whose Collatz trajectory has exactly l odd steps and k even steps arranged in a valid (OO-free) order. The mediant tree does exactly that: it simultaneously produces the rational approximant and the parity sequence that realises it, as a byproduct of the same iterative construction.

More broadly, the mediant tree is a general-purpose encoder: given an oracle for any real number, it produces a parity sequence over $\{\mathsf{O}, \mathsf{E}\}$ that encodes it to arbitrary precision — the Collatz application is one instance of a universal construction.

References

[1] S. Eliahou, *The $3x+1$ problem: new lower bounds on nontrivial cycle lengths*, Discrete Mathematics **118** (1993), 45–56.

A Proof of Theorem 4.1

We derive the closed-form formula for the real-valued minimiser of the combined approximation error $\max(|\varepsilon_j|, |\varepsilon_k|)$.

We have a bounded pair $(r_i, r_{i+1}) = (e_i/o_i,\, e_{i+1}/o_{i+1})$ bracketing z, with $|\varepsilon_i| > |\varepsilon_{i+1}|$. The two weighted mediants are

$$r_j = \frac{e_i + (w-1)\, e_{i+1}}{o_i + (w-1)\, o_{i+1}}, \qquad r_k = \frac{e_i + w\, e_{i+1}}{o_i + w\, o_{i+1}}.$$

Write $O_j = o_i + (w-1)\, o_{i+1}$ and $O_k = o_i + w\, o_{i+1}$ for their denominators.

Step 1: Express ε_j and ε_k as functions of w.

$$\varepsilon_j = \frac{e_i + (w-1)e_{i+1}}{z\, O_j} - 1, \qquad \varepsilon_k = \frac{e_i + we_{i+1}}{z\, O_k} - 1.$$

Define the shorthand

$$A = e_i - z\, o_i, \qquad B = e_{i+1} - z\, o_{i+1}.$$

Note that $A = z\, o_i\, \varepsilon_i$ and $B = z\, o_{i+1}\, \varepsilon_{i+1}$: they are the signed absolute errors of r_i and r_{i+1}, scaled by their respective denominators and by z. With this shorthand the numerators simplify:

$$e_i + (w-1)\, e_{i+1} - z\, O_j = (e_i - z\, o_i) + (w-1)(e_{i+1} - z\, o_{i+1}) = A + (w-1)B,$$

giving

$$\varepsilon_j = \frac{A + (w-1)B}{z\, O_j}, \qquad \varepsilon_k = \frac{A + wB}{z\, O_k}.$$

Step 2: Relate A, B to Δ and K. Recall $\Delta = e_{i+1}o_i - e_i o_{i+1}$ and $K = o_{i+1}(e_{i+1} - z o_{i+1}) = o_{i+1}B$, so immediately $B = K/o_{i+1}$. For A, compute:

$$A\, o_{i+1} - B\, o_i = (e_i - z o_i)o_{i+1} - (e_{i+1} - z o_{i+1})o_i = e_i o_{i+1} - e_{i+1} o_i = -\Delta,$$

hence $A = (Bo_i - \Delta)/o_{i+1}$. Substituting $B = K/o_{i+1}$:

$$A + (w-1)B = \frac{K(o_i/o_{i+1} + w - 1) - \Delta}{o_{i+1}},$$
$$A + wB = \frac{K(o_i/o_{i+1} + w) - \Delta}{o_{i+1}}.$$

A key identity used in Step 3: for any $O = o_i + c\, o_{i+1}$,

$$B\, O - o_{i+1}(A + cB) = B\, o_i + B\, c\, o_{i+1} - o_{i+1}\, A - c\, o_{i+1}\, B = B\, o_i - o_{i+1}\, A = \Delta.$$

In particular, $B\, O_j - o_{i+1}(A + (w-1)B) = \Delta$ and $B\, O_k - o_{i+1}(A + wB) = \Delta$.

Step 3: Set up the minimisation. Since r_j and r_k straddle z, their errors have opposite signs: $\varepsilon_j \varepsilon_k \leq 0$. Minimising $\max(|\varepsilon_j|, |\varepsilon_k|)$ over real w is equivalent to minimising $|\varepsilon_j \varepsilon_k|$, which (since the product is non-positive) means finding the w at which $\varepsilon_j \varepsilon_k$ is **maximised** (i.e. closest to zero from below). We set $\frac{d}{dw}(\varepsilon_j \varepsilon_k) = 0$.

Differentiating ε_j via the quotient rule. Write $\varepsilon_j = N_j / D_j$ where

$$N_j = \frac{A + (w-1)B}{z}, \qquad D_j = O_j = o_i + (w-1)\, o_{i+1}.$$

Both depend on w, with derivatives:

$$\frac{dN_j}{dw} = \frac{B}{z}, \qquad \frac{dD_j}{dw} = o_{i+1}.$$

Applying the quotient rule $d(N/D)/dw = (N'D - ND')/D^2$:

$$\frac{d\varepsilon_j}{dw} = \frac{(B/z)\, O_j - \big((A + (w-1)B)/z\big)\, o_{i+1}}{O_j^2} = \frac{B\, O_j - o_{i+1}(A + (w-1)B)}{z\, O_j^2}.$$

Applying the identity $B\, O_j - o_{i+1}(A + (w-1)B) = \Delta$ from Step 2:

$$\frac{d\varepsilon_j}{dw} = \frac{\Delta}{z\, O_j^2}.$$

The same calculation for $\varepsilon_k = (A + wB)/(z\, O_k)$ gives:

$$\frac{d\varepsilon_k}{dw} = \frac{B\, O_k - o_{i+1}(A + wB)}{z\, O_k^2} = \frac{\Delta}{z\, O_k^2}.$$

Both derivatives share the same numerator Δ — a direct consequence of the identity in Step 2, and the reason the product-rule expression simplifies cleanly.

Applying the product rule. By the product rule:

$$\frac{d}{dw}(\varepsilon_j\, \varepsilon_k) = \frac{\Delta}{z\, O_j^2} \cdot \varepsilon_k + \varepsilon_j \cdot \frac{\Delta}{z\, O_k^2} = 0.$$

Dividing through by Δ / z (nonzero since $|\Delta| = 1$):

$$\frac{\varepsilon_k}{O_j^2} + \frac{\varepsilon_j}{O_k^2} = 0 \qquad \Longleftrightarrow \qquad \varepsilon_k\, O_k^2 = -\varepsilon_j\, O_j^2,$$

which (since ε_j and ε_k have opposite signs) means $|\varepsilon_k|\, O_k^2 = |\varepsilon_j|\, O_j^2$: the two errors are equal when weighted by the square of their denominator.

Step 4: Solve the quadratic. We need to solve $\varepsilon_k\, O_k^2 + \varepsilon_j\, O_j^2 = 0$. Substituting from Steps 1 and 2:

$$\frac{A + wB}{z\, O_k} \cdot O_k^2 + \frac{A + (w-1)B}{z\, O_j} \cdot O_j^2 = 0 \quad \Rightarrow \quad (A + wB)\, O_k + (A + (w-1)B)\, O_j = 0.$$

Expanding using $O_j = o_i + (w-1)o_{i+1}$, $O_k = o_i + wo_{i+1}$, $A = (Bo_i - \Delta)/o_{i+1}$, and $B = K/o_{i+1}$, then clearing o_{i+1} denominators, one obtains a quadratic in w. Applying the quadratic formula and simplifying (using $|\Delta| = 1$ from Definition 2.4) yields the two real roots:

$$w = \frac{1}{2} - \frac{o_i}{o_{i+1}} + \frac{\Delta \pm \sqrt{\Delta^2 - K^2}}{2K},$$

completing the proof of Theorem 4.1.

Discriminant and termination. The discriminant $\Delta^2 - K^2 = 1 - K^2 \geq 0$ (using $|\Delta| = 1$ from Definition 2.4) since $|K| \leq 1$ throughout the iteration.

- When $K = 0$: r_{i+1} is an exact representation of z; no further improvement is possible and the algorithm terminates.
- When $|K| = 1$: the two roots coincide, giving a unique optimal weight $w = \frac{1}{2} - o_i/o_{i+1} + \Delta/K$.
- Otherwise: two real candidates are produced; the correct integer w is found by taking the floor or ceiling of each and selecting the one that keeps the new pair bracketing z.